SeQRity VPN

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1 Introduction

Our project is based on an idea: making online security feel straightforward even for people who are not directly informed or interested about cybersecurity. As users of public networks during travel, we've experienced the feeling of having to choose between convenience and protection. The need for secure internet access can be found everywhere, including in hospitality settings, but the solutions are either too technical, too resource-expensive, or they place all the burden exclusively on the travellers.

To achieve the most user-friendly solution, we didn't want to build another VPN service as the already existing ones. Instead, we based our idea on a QR-based, plug-and-play experience, with no installation required, to focus on the immediate interaction between people and their surroundings in hotels, B&Bs, and similar structures. With our proposal we tried to combine the intuitive gesture of scanning a QR code with the immediate outcome of protected browsing, in order to explore the value proposition we can deliver to hospitality providers.

When first thinking about our product name, we thought about "*QRVPN*", but during our market analysis we came across another application using the same name, offering a similar service but with a different target, as reported in the Existing Alternative section. We then opted for a wordplay between the terms "*secure*", "*QR*" and "*VPN*".

2 Customer Segments and Early Adopters

Our customers are hotels and bed and breakfasts that want to offer better digital security to their guests without complex technical solutions. We sell to hospitality properties of different sizes from small bed and breakfasts to medium-big hotels or hostels. These hospitality infrastructures typically lack proper secure internet connections, and customers who worry about using public wifi networks may choose competitors with better security instead of their service. Our customers usually want to improve their digital offerings and those who desire an advanced yet simple cybersecurity infrastructure may improve guest satisfaction while standing out from competitors.

Main early adopters are small bed and breakfasts because they are easier and faster to work with compared to larger hotels. These B&Bs typically have simpler network infrastructure that makes installation quicker and less expensive, fewer decisions to make which speeds up the whole process, and owners who are directly involved in daily operations. Early adopters are mainly composed of emerging B&B owners who wish to improve guest satisfaction and provide a unique service.

3 Problem Statement

Nowadays, most travellers rely on public or semi-public Wi-Fi networks offered by hotels, B&Bs, hostels and other accommodation and hospitality providers to connect to the Internet. Especially for international travellers, relying only on mobile data can be costly or limited, making the previously mentioned kind of networks the default option. While convenient and usually free, these shared networks often lack basic security measures, leaving users vulnerable to data interception, tracking and other types of cyber-threats, not avoidable with the use of the password alone, given to all customers at check-in. This issue is particularly critical for guests who access sensitive information such as banking apps or work emails, while being unaware of the risks. Even more aware customers may often lack the tools, knowledge or time to protect themselves effectively.

While travelling, guests expect comfort and safety not only in their physical environment, but also in their digital experience. However, the ever-growing complexity of cybersecurity and the lack of user-friendly solutions, often force guest to choose convenience over safety. For example, most people are not willing or not able to install VPNs, configure their settings, or deal with other technical barriers, especially while on vacation. As a result, they prefer to take the risk of using unprotected networks.

The core need, from the hospitality structures point of view, is to offer a sense of digital security to their guests. They want to ensure a safe, seamless digital experience for their clients, but they currently lack simple and reliable ways to do so. Physical comfort and service quality are often prioritized by most establishments, but digital safety is

becoming a key aspect of the guests' experience, especially as more people travel with sensitive data on their devices.
Clients expect to connect easily to the internet without worrying about privacy breaches, data leaks or account compromise and, if their digital experience feels unsafe, complex or frustrating, it can affect their overall perception of the stay, and this can compromise the reputation of the hospitality provider. This is especially true and important for international travellers, remote workers and business guests, who often depend on secure connectivity during their trips.
From the perspective of accommodation providers, the major problems are:

- **Lack of trust and control:** Providers may be aware that their Wi-Fi may not be fully secure, but upgrading infrastructures or implementing complex cybersecurity solutions is often expensive and technically demanding. This leaves them with limited means to offer guaranteed protection to guests. On the other hand, users are unsure whether the network they're connecting to is safe and this generates anxiety and a constant feeling of being exposed.

- **Negative impact on guest satisfaction:** Guests might feel unsafe or complain about the unreliability of the network. Even if these concerns are not always said out loud, they may even subconsciously affect online reviews or future bookings (which is an increasing concern in a competitive market like this one).

- **Limited differentiation and added value:** Most accommodation providers offer Wi-Fi access as a basic service, but very few use it as a trust-building asset. Inability to offer secure and easy connectivity means missing an opportunity to stand out from other similar establishments by addressing a rising concern among travellers.

In short, the problem is the gap between guests' expectations of safe internet access and the insecure reality of shared networks, due to the current limitations of hospitality services. Our goal is to reduce this gap by helping providers offer a solution that makes digital safety as accessible and intuitive as logging into Wi-Fi in the more conventional way.

4 Existing Alternatives

When hospitality establishments want to offer safe internet access to their guests, they typically face fragmented and limited options. In most cases, they provide access to a shared Wi-Fi network secured by a generic password: this approach is convenient in the short-term, but does not protect guests' traffic. As cybersecurity awareness increases among travellers, some establishments are looking for better ways to enhance their network safety, but often use partial solutions that do not fully address the problem.

One common workaround is the use of legal disclaimers or terms of service that warn guests about the lack of responsibility for data security. While this option might protect the hospitality providers from liability, it does not provide any real solution to the guests' need for a secure connection. Some structures try to upgrade their hardware using professional routers or captive portals, but these options often require significant technical investment and do not always improve the user experience in the end.

From the guests' side, those who are more aware of digital risks may try to protect themselves by using the approaches presented before, like using the mobile data, with the relative limitations already discussed. Others may install VPN apps manually, but this process requires time, digital knowledge and even paid subscriptions in some cases, barriers that are often seen as overwhelming by many casual users, especially during short stays.

While hospitality venues do not currently have a simple, plug-and-play solution to solve the problem with minimal effort, and most simply hope that no incidents occur, some companies are starting to address this issue either directly or indirectly. Among the direct competitors, we can find providers like NordLayer and OpenVPN Cloud. These platforms offer VPN and network protection services for the broader target of organizations and could be adapted for hospitality use, but they are primarily designed for corporate environments and are not optimized for the more specific hospitality sector in term of user interface or guest experience.

NordLayer [1], previously known as NordVPN Teams, offers a cloud-based business VPN that allows companies to manage employee access, monitor usage, and enforce security policies across distributed teams. It includes a dedicated IP service, threat-

blocking tools and network segmentation features. However, it requires user login and device management, making it unsuitable for casual, short-term hotel guests.
CloudConnexa [2], formerly OpenVPN Cloud, provides secure tunneling and centralized management of encrypted connections, enabling companies to configure multiple access points and monitor traffic. Its setup demands technical expertise and assumes a level of IT control that most hospitality venues do not have.
A notable competitor is **Qrvpn** [3] by VentoByte, which enables users to deploy VPN access via QR codes. While being an interesting alternative with a really similar concept, Qrvpn uses a similar technology but focusing on individual or enterprise users, rather than hospitality use cases. It does not offer a smooth and easy experience for non-technical hotel guests, nor does it integrate into a managed service for the accomodation provider. Although it supports multiple platforms such as Android, iOS, Windows, and Linux, it still requires users to download an application and manage the VPN configuration themselves. Moreover, the description provided on the Google Play Store [4] highlights its target audience, suggesting that the app is primarily intended to allow users to securely access private networks they already control, rather than providing a public-facing and guest-friendly solution for our target environment. As such, Qrvpn serves a different use case and user profile than the one targeted by our solution.

Other indirect competitors fulfill the same customer need but through alternative means or for different sectors. For instance, ProtonVPN, Surfshark, ExpressVPN, and similar consumer VPN brands allow individual travellers to protect their own devices. However, they operate independently from the hospitality structure and are not always accessible or user-friendly for short-term users. These solutions serve the same underlying need, but through a self-managed, guest-initiated approach, which limits their practicality in our context:
ProtonVPN [5] focuses on privacy-first policies and offers a free tier, but requires users to create an account and install the app, which may stop casual guests;
Surfshark [6] offers affordable multi-device protection and extra services like ad-blocking, but still requires manual setup and does not integrate with the hospitality venue's network and experience;
ExpressVPN [7] is known for its user-friendly interface and simple configuration,

but it's a premium product with a subscription model, impractical for guests staying only a few days (and mostly impractical for hotels, B&Bs and hostels), since it offers a free-trial, but it is of course non-repeatable.

Cloudflare for Teams [8] focuses on DNS-level filtering. While being well-suited for users in an enterprise network, it lacks a component thought particularly for guests usage.
Some technically skilled guests may use SSH tunneling, Tor, or proxy chaining to stay secure online, so we can see these methods as indirect competitors as well, but they are far too complex to be considered viable alternatives for the average hotel guest or for scalable implementation by hospitality businesses.
Even more indirect solutions include mobile hotspots like Solis [9], which let travelers bypass local Wi-Fi completely by using independent 4G/5G connections. However, these are expensive, limited to specific regions, and again do not involve the hospitality venue in solving the guest's security concerns, all the problems previously addressed.
So, to summarize, while there are several ways to address the issue of unsafe Wi-Fi in hospitality environments, no current solution satisfies the combined needs of both guests and venues, since the existing options either shift the burden onto the guest, require high and expensive technical investment, or neglect the user experience. This leaves a significant opportunity for a solution that is secure, simple and tailored specifically to out target context.

5 Solution

SeQRity VPN solves frequent cyberattacks that happen on public networks in hotels and bed and breakfasts with the simplest approach possible: scanning a QR code that enables secure VPN connection directly through the hotel infrastructure. We install a custom firmware into hotel or B&Bs routers that enables protected network segments accessible only through QR codes placed inside the rooms, eliminating the need for guests to download apps or configure complex security settings.
The guests experience is the effortless. They pay a small fee upon check-in in order to receive a qr-code inside their room, when they scan the code they automatically

connect to the structure WIFI and receive a secure encrypted connection they can use throughout their entire stay.
The hotels and B&Bs are incentivized to offer this service since it directly enhances their reputation and addresses growing guest concerns about credit card theft and data breaches in hospitality environments. The properties benefit also from a point-based reward system where they earn more points as guests request and use the service, these points can be later spent to obtain prizes like Amazon gift card or smartboxes. As an indirect service they also receive technical support for their network infrastructure, which often lacks proper cybersecurity measures, this enables us to access to usage metrics in order to keep track of user satisfaction.
This solution directly addresses the three main customer frustrations. First, user usually dislikes downloading apps, creating accounts or remembering password therefore guests only need to scan a QR code instead, this keeps the approach to the service as simple as possible. Secondly, all guest traffic is encrypted using our vpn, preventing data theft and credit card fraud that commonly occurs on unsecured public networks such as hotels or B&Bs. Lastly, connectivity problems are eliminated because our optimized network provide faster, more reliable internet access compared to slow third-party VPN applications that guests typically install on their devices, or the need to use mobile data connection.
SeQRity VPN service requires partnerships and faces important limitations. We partner with three essential partnerships

- **Router manufacturers**: We need direct collaboration with router manufacturers because our solution requires installing custom firmware on their devices. These partnerships ensure hardware compatibility and allow us to develop firmware that integrates with their existing router and infrastructure.

- **Cloud infrastructure providers**: Our service depends on cloud providers to host VPN servers in multiple geographic locations. Without these, we cannot provide the encrypted tunnels that protect guest data or ensure low-latency connections for guests

- **Hotels**: Unlike individual bed and breakfasts, hotels provide the network infrastructure scale and standardization we need to deploy our solution efficiently.

An uniform router configuration across properties enable better management and simplify maintenance of our custom firmware installations.

The main limitation is that hotels and B&Bs become dependent on our technical infrastructure since security systems are built into their network hardware. If our service experiences downtime, guests cannot access secure connections until systems are restored. Additionally, older hotels or B&Bs may require expensive router upgrades, and some corporate travelers might experience conflicts between our router-level security and their company VPN requirements.

6 Unfair Advantage

Our SeQRity VPN offers a unique combination of low-effort user experience and deep integration with the operational needs of hotels and B&Bs. Unlike existing alternatives, it requires no app installation, no user account (from the point of view of the guests), and no technical setup after the first general network configuration and the installation of the infrastructure. The only thing needed is a quick QR scan. This makes it instantly accessible, even for the average user. SeQRity VPN is imagined with a focus on hospitality providers, and for them it adds value by enhancing guest trust, modernizing their digital offering, and improving reputation. Even from an emotional point of view, it creates a sense of care and reliability: guests feel protected without having to ask. It bridges the gap between unmanaged public Wi-Fi and complex enterprise tools by offering what is perceived by the clients of the hotels and B&Bs as something similar to a plug-and-play security, designed specifically for the hospitality context. No other competitor combines privacy, simplicity, and operational integration in a single experience, using immediate tools like QR codes.

7 Unique Value Proposition

"SeQRity VPN, your complete WiFi solution: protect guests, boost reputation, earn rewards with no IT headaches."

This message speaks directly to what hotels managers and B&Bs hosts care about most. We start with "complete WiFi solution" because hospitality properties manager don't want another complicated system, they want something that works and solves their problem entirely. "Protect guests" addresses their main worry about data breaches and unhappy customers who get hacked on their network. "Boost reputation" is a direct consequence on offering a reliable service, in the era of increasing cybersecurity threats over public networks. "Earn rewards" speaks directly to managers or hosts that actually get something back through our points system. Most importantly, "no IT headaches" promises what every hotel manager wants to hear, this won't create more work. Our message covers all aspects of our service: the security benefit for guests, the business advantages for properties, the reward incentives, and the simplicity that makes it all possible. By presenting ourselves as the complete solution, we show that hospitality structures don't need different complex services or worry about technical details to ensure a secure connection, just install our firmware once and everything works automatically.

8 High-Level Concept

"Keep It Simple, Keep It SeQR - Shielded navigation through a single QR scan."

We have chosen this slogan since it perfectly captures SeQRity VPN's essence by combining memorable wordplay with our core promise. "Keep It Simple, Keep It SeQR" creates an instantly recognizable phrase that hotels can use in their marketing while implementing our service. The repetitive sound makes it easy to remember, functioning as both a promise and instruction, it explains what our service do and how to use it. "Shielded navigation" it's a simple concept that explains the utility of using a VPN without repeating "secure", guests understand they'll browse safely without needing to know about encryption or VPN protocols. "Through a single QR scan" emphasizes our key differentiator: while competitors require app downloads, account creation, and complex configurations instead we "keep it simple". Our slogan works for both our customers and their guests since it's simple enough that everyone can grasp the underlying meaning of it. By transforming cybersecurity from a complex challenge into a simple scan, we've created a new standard for hospitality WiFi that prioritizes both protection and user experience.

9 Key Metrics

For measuring the success of our company and evaluating our market performance inside hotels and bed and breakfasts, we will track and analyze five key metrics that will help us to track our key performance indicators:

- **QR code conversion rate**: Percentage of guests who request the QR service at check-in and then actually scan and use the QR codes in their rooms to measure the effective success of our service.
- **Customer satisfaction rate**: Feedback from both guests and hotel/BnBs regarding simplicity of use, network performance and perceived usefulness of the service.
- **Hotels and B&Bs adoption rate**: Number of new structures installing our service each month to measure market demand and show how much hotel and bed and breakfasts are willing to invest in our service.
- **Network data usage metrics**: Total amount of secure traffic and connection time through our custom firmware to understand how much guests actually use our protected network during their stay.
- **Customer acquisition cost efficiency**: Cost required to acquire each new hotel/Bed and breakfast partnership compared to the revenue they generate to ensure our business model remains profitable.

10 Cost Structure

Our company requires various resources and investments to deliver our cybersecurity service to the hospitality market, particularly subscription and labor costs:

- **Initial installation costs**: Expenses for deploying custom firmware technology in hotel and BnB networks and adapting the infrastructure, this may be considerable if the infrastructure is outdated or located in an isolated position.

- **Maintenance costs**: Expenses for providing technical support, firmware updates and dashboard updates to ensure performant internet connection and reliable security. These costs should be contained over the years.
- **Cloud infrastructure costs**: Subscriptions for hosting VPN servers, partners may be needed to ensure widespread server locations at affordable prices with reliable bandwidth. This may be the most costly part of the service, therefore we may plan for on-premise solutions in future.
- **Firmware development costs**: Initially custom firmwares must be developed, requiring employed developers with correct tools and training. Since this is a cybersecurity product, frequent updates must be released to ensure safety. Developing custom products can be expensive initially, updates should cost less.
- **Communication costs**: Expenses for promoting our services to hotels and BnBs, including digital marketing campaigns. Low expenses are expected for this area.
- **Prize costs**: Expenses for rewarding hotels and B&Bs with our point system. They may include Amazon gift cards, smartboxes or other type of prizes. Costs are negligible since this promotion should incentivize the usage of our product.

11 Revenue Streams

Our company generates revenue through tiered subscription packages targeting different segments of the hospitality infrastructures market:

- **Basic tier subscription**: Basic monthly package for small BnBs including fundamental firmware installation, limited QR codes and basic technical support services designed specifically for small bed and breakfast.
- **Advanced tier subscription**: Comprehensive package including everything from the basic tier plus advanced firmware features, significantly increased number of QR codes and a basic analytics dashboard, designed for small and medium size hotels.

- **Ultimate tier subscription**: Premium package including everything from the advanced tier plus custom firmware configurations, unlimited QR codes, dedicated advanced technical support and advanced customizable analytics dashboard. Designed for big size hotels.

- **Pay-per-use QR activation**: Additional revenue from properties exceeding their tier QR code limits, it is possible to pay for more qr if guests demands them.

- **Setup and onboarding fees**: One-time payments for initial infrastructure installation, network configuration of the chosen subscription tier.

12 Conclusion

The idea presented in this report is potentially feasible in theory, but it presents some limitations that make it challenging to implement in real-world scenarios, especially for small-scale entrepreneurs. The major complication in our opinion would be that it requires an initial investment that is far too high-cost. Secondly, for hotel managers and B&B hosts that have never dealt with, or maybe even thought about, digital security, it's unlikely to see it as a priority. In such cases, it becomes difficult to convince them to invest time, money, and attention into a solution for a problem they don't recognize as relevant. For these reasons, we can expect a low likelihood of full success or immediate profit. The perceived risk may exceed the perceived benefits from an investor's point of view. However, from a user perspective, we think that having this solution available, already implemented, and ready to use, would potentially be very useful, as it would provide a cheaper and reliable alternative to other third-party VPNs, adding value.
To conclude, we would personally use the service presented if already available, but we would feel held back by the limitations regarding a direct investment of resources.

References

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